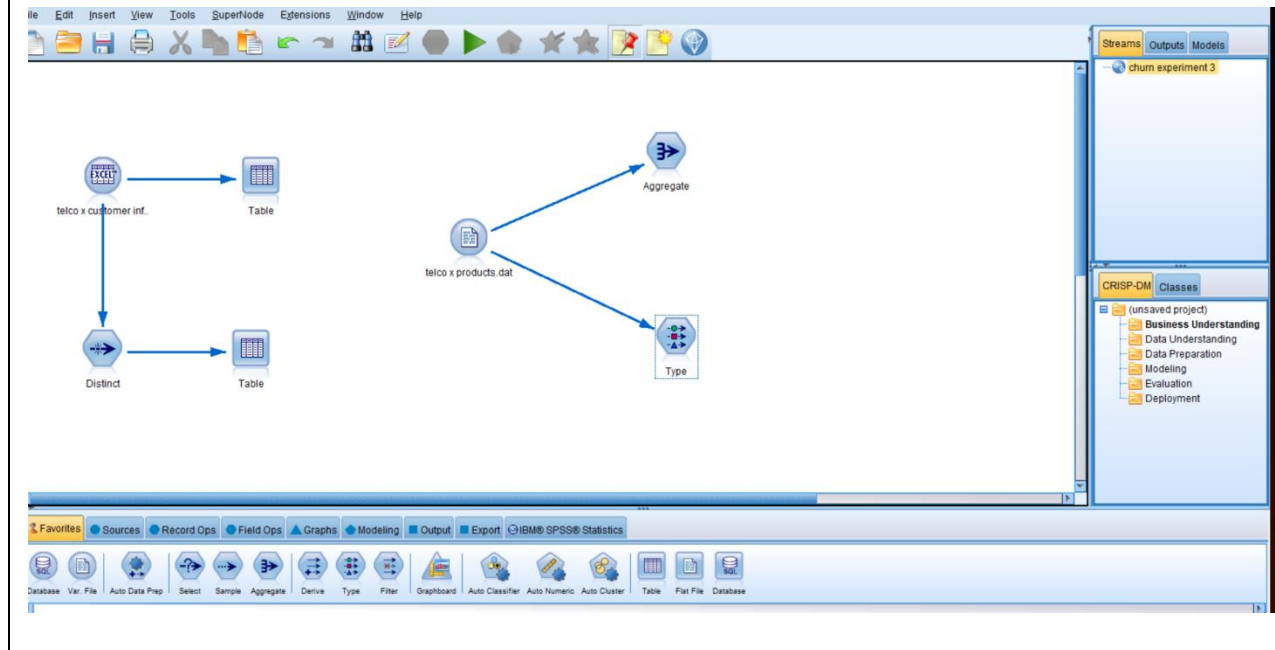


ASSIGNMENT-3



Step 1: Load Customer Information Data

1. Open **IBM SPSS Modeler**.
2. From the **Sources** palette, drag an **Excel** node into the stream.
3. Load the file “**telco x customer info**”.

4. Connect the Excel node to a **Table** node.
 5. Run the Table node to view and verify the customer data.
-

Step 2: Remove Duplicate Customer Records

1. From **Record Ops**, drag a **Distinct** node into the stream.
 2. Connect the **Excel (customer info)** node to the
 3. Configure the Distinct node using **Customer ID** as the key field.
 4. Connect the Distinct node to a **Table** node.
 5. Run the Table node to confirm duplicates have been removed.
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Step 3: Load Product Data

1. From **Sources**, drag a **Flat File** node into the stream.
 2. Load the file “**telco x products.dat**”.
 3. Run the node to ensure the data loads correctly.
-

Step 4: Define Field Measurement Levels

1. From **Field Ops**, drag a **Type** node.
 2. Connect the **telco x products.dat** node to the **Type** node.
 3. Set appropriate measurement levels:
 - Nominal: Customer ID, Product Type
 - Continuous: Charges, Usage, Tenure
 - Flag/Target: Churn
 4. Execute the Type node to Apply Change.
-

Step 5: Aggregate Product Data

1. From **Record Ops**, drag an **Aggregate** node.

2. Connect the **telco x products.dat** node to the **Aggregate** node.
 3. Group by **Customer ID**.
 4. Apply aggregation functions such as:
 - Total charges
 - Average usage
 - Count of products
 5. Run the Aggregate node to generate summarized customer-level data.
-

Step 6: Validate the Output

1. Attach a **Table** node to the Aggregate output.
 2. Run the Table node.
 3. Verify that aggregation and field types are correctly applied.
-

Step 7: Prepare for Modeling

1. Save the stream as “**churn experiment 3**”.

2. Ensure all data preparation steps are complete.
 3. The dataset is now ready for churn modeling using classification techniques.
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